Γ-point head-and-wing correction for Γ-only real-space RI *GW* of two- and three-dimensional periodic systems

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Large-cell *GW* calculations of defects, interfaces and moiré superlattices are naturally done at the Γ point of a supercell, but the screened interaction W at the single sampled wavevector $\mathbf{q} = 0$ is ill defined: the real-space lattice sum of the bare Coulomb operator between charged auxiliary functions diverges there. We present a scheme that makes Γ-only *GW* with real-space resolution of the identity (RI-RS) well defined for two-dimensional slabs and three-dimensional bulk cells. The divergence is confined to a single direction of the auxiliary basis, the charge channel. In two dimensions it is removed from $V(0)$ as an exactly computable rank-one constant; in three dimensions $V(0)$ is taken as an Ewald sum with the $\mathbf{G} = 0$ term omitted. The charge channel is then carried analytically by one additional basis function, the normalized constant, whose screened head and wings are obtained in closed form from velocity-gauge transition dipoles and the Gaussian dielectric body and are integrated over the whole small Brillouin zone of the supercell. Nothing is subtracted, because the single sampled point carries no head once the constant has been removed. In two dimensions the correction is applied to W in the auxiliary basis; in three dimensions, where the head is eV-sized, it is applied to the correlation self-energy in the orbital basis through a single orbital-pair kernel that contains the head, the head-dipole and the wing terms. We give every working equation in the order of evaluation, describe the implementation in CP2K, and discuss why the earlier k-mesh large-cell scheme is not a valid reference for two-dimensional systems.

I. INTRODUCTION

Many-body perturbation theory in the *GW* approximation is the standard route to quasiparticle band structures of solids, surfaces and two-dimensional (2D) materials. For systems that need large simulation cells — point defects, heterostructures, moiré superlattices with thousands of atoms — the Brillouin zone of the supercell is small, and it is natural to evaluate the polarizability, the dielectric function and the screened interaction at a single wavevector, the Γ point of the supercell. The real-space resolution-of-the-identity (RI-RS) formulation of *GW* in CP2K is designed for this regime: the Green's function and the polarizability are element-wise products on an atom-centred real-space grid, every contraction with the auxiliary basis goes through sphere-local fit coefficients, and the cost grows with the number of atoms rather than with the number of k-points.

The large-cell code in CP2K reconstructs $\chi(\mathbf{k})$, $\varepsilon(\mathbf{k})$ and $W(\mathbf{k})$ from Γ on a k-mesh `KPOINTS_W` $n_1 n_2 n_3$ of the supercell's own small Brillouin zone and takes the bare Coulomb matrix $V(\mathbf{k})$ from a finite real-space image sum. An even mesh is half-shifted and never samples $\mathbf{q} = 0$; an odd mesh does. The obvious Γ-only setting, `KPOINTS_W 1 1 1`, therefore samples exactly the wavevector at which the image sum of the bare Coulomb operator between two charged auxiliary Gaussians diverges with the summation range. It does not merely sample badly: it produces a large spurious additive contribution to W. On a 392-atom hexagonal boron nitride (h-BN) slab the correlation self-energy Σ^c of the valence band maximum moved from $+2.195$ to -24.043 eV and that of the conduction band minimum from -3.283 to $+20.237$ eV (opposite signs, because $\Sigma^c = -GW$ and the two branches of G carry opposite sign), while the exchange self-energy Σ^x and v^{xc} stayed bit-identical.

This work makes `KPOINTS_W 1 1 1` well defined. The divergence is shown to sit in one direction of the auxiliary basis, the charge channel, which is removed from the sampled $V(0)$ and put back analytically through one additional basis function, the normalized constant. Its contribution to the correlation part of the screened interaction (the "head" and the "wings") is known in closed form as a function of $\mathbf{q}$ and is integrated over the whole small Brillouin zone of the supercell, following the head-and-wing construction of J. Wilhelm for full-k RI-RS *GW* (Ref. 1, Sec. VI; CP2K PR #6447 [2]). Because the regular path is Γ only and its sampled head is removed exactly, nothing has to be subtracted.

Section II gives every equation of the scheme for a 2D slab in the order of evaluation; Sec. III gives the changes for 3D bulk cells, where the Coulomb matrix is an Ewald sum and the correction is applied in the orbital basis. Section IV describes the implementation in CP2K, Sec. V the results, and Sec. VI discusses why the old k-mesh large-cell scheme is not a reference for 2D systems and why Gaussian form factors of the auxiliary charges can be neglected.

II. THEORY FOR TWO-DIMENSIONAL SLABS

A. Setting and notation

The simulation cell is a layer with in-plane lattice vectors $\mathbf{a}_1, \mathbf{a}_2$, periodic in x and y only, with a non-periodic direction z. Its layer area is

$$A = |\mathbf{a}_1 \times \mathbf{a}_2|, \tag{1}$$

and the in-plane reciprocal vectors $\mathbf{b}_1, \mathbf{b}_2$ satisfy $\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi\delta_{ij}$. The vacuum height never enters any equation below: all Fourier transforms average over a layer cell, not over a three-dimensional cell that mostly contains vacuum.

Indices: $\mu, \nu, \lambda, \sigma$ label atomic orbitals (AOs) φ_μ; P, Q, S label auxiliary (RI) Gaussians ϕ_P; ℓ, ℓ' label points $\mathbf{r}_\ell$ of a real-space grid; i labels occupied and a empty Kohn–Sham states;

n,m label any state; $\mathbf{R}$ labels lattice vectors; τ is imaginary time and ω imaginary frequency. $\alpha,\beta \in \{x,y,z\}$ are Cartesian directions and $\hat{\mathbf{q}}$ an in-plane unit vector. All quantities are real: there is one real k-point, Γ, and the single complex factor of the construction (the wing of W) is split off analytically.

The RI metric is a short-ranged kernel v_ω, the truncated Coulomb operator $v_{\mathrm{tc}}(r) = \theta(r_c - r)/r$ in the calculations reported here. Its zero-wavevector value is

$$\int \mathrm{d}\mathbf{r}\, v_\omega(r) = \begin{cases} \pi/\omega^2 & \text{erfc-attenuated metric,} \\ 2\pi r_c^2 & \text{truncated Coulomb metric,} \end{cases} \tag{2}$$

and is referred to as the metric head.

Imaginary time and frequency are discretized on minimax grids $\{\tau_j\}$, $\{\omega_k\}$, $j,k = 1,\ldots,N$. The cosine and sine transforms are written with their minimax weights,

$$\begin{aligned} f(\mathrm{i}\omega_k) &= \sum_j \gamma^{\cos}_{kj} f(\mathrm{i}\tau_j), \qquad f(\mathrm{i}\tau_j) = \sum_k \bar{\gamma}^{\cos}_{jk} f(\mathrm{i}\omega_k), \\ f(\mathrm{i}\omega_k) &= \sum_j \gamma^{\sin}_{kj} f(\mathrm{i}\tau_j), \end{aligned} \tag{3}$$

where the first represents $f(\mathrm{i}\omega) = 2\int_0^\infty \mathrm{d}\tau \cos(\omega\tau) f(\mathrm{i}\tau)$ for an even function of τ, the second its inverse, and the third $f(\mathrm{i}\omega) = 2\mathrm{i}\int_0^\infty \mathrm{d}\tau \sin(\omega\tau) f(\mathrm{i}\tau)$ for an odd one.

B. Atomic orbitals on the real-space grid

Modified Lebedev atomic grids are placed at every atom and their union is the global grid $\{\mathbf{r}_\ell\}_{\ell=1}^{N_{\mathrm{grid}}}$, all points in the home cell. Because there is only one k-point, the AO values on the grid are the Γ Bloch sums over periodic images,

$$\Phi_\mu(\mathbf{r}_\ell) = \sum_{\mathbf{R}} \varphi_\mu(\mathbf{r}_\ell - \mathbf{A}_\mu - \mathbf{R}), \tag{4}$$

where $\mathbf{A}_\mu$ is the center of φ_μ. The matrix $\Phi_{\ell\mu}$ is sparse: $\Phi_{\ell\mu}$ is kept only if $|\mathbf{r}_\ell - \mathbf{A}_\mu - \mathbf{R}|$ is within the AO radius for some image. At the 14×14 h-BN cell 24.9 % of its elements are nonzero.

C. Real-space RI: the fit coefficients $Z_{\ell P}$

RI-RS expresses products of AOs through the grid. Define the grid metric and the grid–RI right-hand side,

$$D_{\ell\ell'} = \Big[\sum_\mu \Phi_\mu(\mathbf{r}_\ell)\Phi_\mu(\mathbf{r}_{\ell'})\Big]^2, \tag{5}$$

$$d_{\ell P} = \sum_{\mu\nu} \Phi_\mu(\mathbf{r}_\ell)\Phi_\nu(\mathbf{r}_\ell)\,(\mu\nu|P)_\omega, \tag{6}$$

with

$$(\mu\nu|P)_\omega = \sum_{\mathbf{R}\mathbf{R}'} \iint \mathrm{d}\mathbf{r}\,\mathrm{d}\mathbf{r}'\, \varphi^{\mathbf{R}}_\mu(\mathbf{r})\varphi^{\mathbf{R}'}_\nu(\mathbf{r})\, v_\omega(|\mathbf{r}-\mathbf{r}'|)\,\phi_P(\mathbf{r}'). \tag{7}$$

Writing $B_{(\mu\nu),\ell} = \Phi_\mu(\mathbf{r}_\ell)\Phi_\nu(\mathbf{r}_\ell)$ one has $D = B^{\mathrm{T}}B$ and $d_{\cdot P} = B^{\mathrm{T}} g_P$ with $g_{(\mu\nu),P} = (\mu\nu|P)_\omega$, so the normal equations

$$\sum_{\ell'} D_{\ell\ell'} Z_{\ell' P} = d_{\ell P} \tag{8}$$

are the least-squares solution of $\sum_\ell \Phi_\mu(\mathbf{r}_\ell)\Phi_\nu(\mathbf{r}_\ell) Z_{\ell P} \approx (\mu\nu|P)_\omega$: the coefficient $Z_{\ell P}$ is a quadrature weight times the value at $\mathbf{r}_\ell$ of the metric potential of ϕ_P. The solve is balanced and regularized,

$$\begin{aligned} j_\ell &= \frac{1}{\max(D_{\ell\ell}, \sqrt{\varepsilon_{\mathrm{floor}}})}, \\ \big[j_\ell D_{\ell\ell'} j_{\ell'} + \lambda\,\delta_{\ell\ell'}\big] Z'_{\ell' P} &= j_\ell d_{\ell P}, \qquad Z_{\ell P} = j_\ell Z'_{\ell P}, \end{aligned} \tag{9}$$

with the Tikhonov parameter $\lambda = 10^{-8}$ and $\varepsilon_{\mathrm{floor}} = 10^{-16}$. Because $d_{\ell P}$ decays away from the center of ϕ_P, Eq. (8) is solved independently for each atom P on the grid points inside a sphere around it (radius $r_{\mathrm{RI}} + r_{\mathrm{AO}}$), by a dense Cholesky factorization of the sphere's block of D. This sphere locality is what makes RI-RS linear scaling; it is valid precisely because every column $d_{\ell P}$ is localized, a point that becomes important in Sec. II H 1.

Every subsequent contraction between the grid and the RI basis uses the same matrix:

$$X_{PQ} = \sum_{\ell\ell'} Z_{\ell P} X_{\ell\ell'} Z_{\ell' Q}, \qquad X_{\ell\ell'} = \sum_{PQ} Z_{\ell P} X_{PQ} Z_{\ell' Q}. \tag{10}$$

D. Green's functions and polarizability at Γ

With Kohn–Sham coefficients $C_{\mu n}$, eigenvalues ε_n and the Fermi level ε_F placed mid-gap, the occupied and empty Green's functions in imaginary time are

$$G^{\mathrm{occ}}_{\mu\nu}(\mathrm{i}|\tau|) = \sum_i^{\mathrm{occ}} C_{\mu i}\, \mathrm{e}^{-|(\varepsilon_i - \varepsilon_F)\tau|} C_{\nu i},$$

$$G^{\mathrm{vir}}_{\mu\nu}(\mathrm{i}|\tau|) = \sum_a^{\mathrm{vir}} C_{\mu a}\, \mathrm{e}^{-|(\varepsilon_a - \varepsilon_F)\tau|} C_{\nu a}, \tag{11}$$

$$G^{\mathrm{occ/vir}}_{\ell\ell'}(\mathrm{i}|\tau|) = \sum_{\mu\nu} \Phi_\mu(\mathbf{r}_\ell)\, G^{\mathrm{occ/vir}}_{\mu\nu}(\mathrm{i}|\tau|)\, \Phi_\nu(\mathbf{r}_{\ell'}), \tag{12}$$

each built as $(\tilde{C})(\tilde{C})^{\mathrm{T}}$ with $\tilde{C}_{\mu n} = C_{\mu n}\mathrm{e}^{-|(\varepsilon_n - \varepsilon_F)\tau|/2}$. The independent-particle polarizability is an element-wise (Hadamard) product on the grid, contracted to the RI basis with Eq. (10),

$$\begin{aligned} \chi_{\ell\ell'}(\mathrm{i}\tau) &= G^{\mathrm{occ}}_{\ell\ell'}(\mathrm{i}|\tau|)\, G^{\mathrm{vir}}_{\ell\ell'}(\mathrm{i}|\tau|), \\ \chi_{PQ}(\mathrm{i}\tau) &= \sum_{\ell\ell'} Z_{\ell P}\, \chi_{\ell\ell'}(\mathrm{i}\tau)\, Z_{\ell' Q}. \end{aligned} \tag{13}$$

The grid product is evaluated in panels of grid rows; a geometric cutoff on $|\mathbf{r}_\ell - \mathbf{r}_{\ell'}|$ limits the pairs. At Γ, $\chi_{\ell\ell'}$ is a Gram matrix of occupied–empty products and therefore exactly positive semidefinite; no clipping of negative eigenvalues is needed. It is transformed to imaginary frequency with the cosine weights of Eq. (3), $\chi_{PQ}(\mathrm{i}\omega_k) = \sum_j \gamma^{\cos}_{kj} \chi_{PQ}(\mathrm{i}\tau_j)$.

E. The screened interaction at Γ: the problem and the plan

W is needed at one point only, $\mathbf{q} = 0$. There the bare Coulomb matrix between charged auxiliary functions diverges. The scheme handles this in three steps:

1. The divergence sits in one direction of the auxiliary basis, the charge vector c. It is removed from $V(0)$ as a single constant, $V(0) \to V(0) - D_{\text{box}}\, cc^{\mathrm{T}}$ (Sec. II F).
2. ε and W are computed from this finite $V(0)$ in the usual way (Sec. II G). What is now missing from W is exactly the screened interaction in the charge channel.
3. That missing piece is added back analytically: the charge channel is represented by the constant function $\phi_0 = 1/\sqrt{A}$, its contribution to W (the "head" and "wings") is known in closed form as a function of $\mathbf{q}$, and it is integrated over the whole small Brillouin zone of the supercell (Sec. II H). Nothing has to be subtracted, because step 1 already removed it from the sampled point.

F. Step 1: the bare Coulomb matrix at $\mathbf{q} = 0$

1. What is computed

The Coulomb matrix between two auxiliary functions ϕ_P and ϕ_Q in a periodic layer is the interaction of ϕ_P in the home cell with ϕ_Q in *every* cell. The code sums over a finite box of cells,

$$V^{\text{lat}}_{PQ}(0) = \sum_{\mathbf{R}\in\mathscr{B}} \underbrace{\iint \mathrm{d}\mathbf{r}\,\mathrm{d}\mathbf{r}'\, \frac{\phi_P(\mathbf{r})\,\phi_Q(\mathbf{r}'-\mathbf{R})}{|\mathbf{r}-\mathbf{r}'|}}_{(P0|Q\mathbf{R})}, \tag{14}$$

$$\mathscr{B} = \{n_1\mathbf{a}_1 + n_2\mathbf{a}_2 : |n_1|, |n_2| \le m\}, \tag{15}$$

with $m = (3^{s-1}-1)/2$ and s the input `SIZE_LATTICE_SUM` ($s = 4$: $m = 13$, i.e. 27×27 cells). There are no images in the direction normal to the layer.

2. Far images see only the charge

Take one term $(P0|Q\mathbf{R})$ with $|\mathbf{R}|$ much larger than the extent of the two Gaussians. Seen from far away, ϕ_Q looks like a point charge. Its charge is its integral,

$$c_Q = \int \mathrm{d}\mathbf{r}\,\phi_Q(\mathbf{r}), \tag{16}$$

and the same for ϕ_P. The multipole expansion of the term is therefore

$$(P0|Q\mathbf{R}) = \frac{c_P c_Q}{|\mathbf{R}|} + \frac{(\text{dipole terms})}{|\mathbf{R}|^2} + \frac{(\ldots)}{|\mathbf{R}|^3} + \cdots \tag{17}$$

Summed over the box, the dipole terms cancel between $\mathbf{R}$ and $-\mathbf{R}$ and the $|\mathbf{R}|^{-3}$ and higher terms converge. The only part of the sum that does not settle down is the first one:

$$V^{\text{lat}}_{PQ}(0) = c_P c_Q \sum_{\mathbf{R}\in\mathscr{B}} \frac{1}{|\mathbf{R}|} + \big(\text{converges as the box grows}\big). \tag{18}$$

3. Why the first part diverges

In two dimensions the number of lattice points in a ring of radius R and width $\mathrm{d}R$ is $2\pi R\,\mathrm{d}R/A$ (one point per area A). Each of them contributes $1/R$, so the ring contributes $2\pi\,\mathrm{d}R/A$ — the same for every ring. The sum therefore grows *linearly* with the box radius and never converges:

$$\sum_{\mathbf{R}\in\mathscr{B}} \frac{1}{|\mathbf{R}|} \;\approx\; \frac{1}{A}\int_{\mathscr{P}} \frac{\mathrm{d}^2 r}{|\mathbf{r}|} \;\propto\; \frac{2\pi}{A} R_{\text{box}}. \tag{19}$$

On the h-BN cell this term is 7.4×10^4 Ha in the charge direction (defined next), against a few Ha for everything else.

4. The charge channel

Collect the charges into one vector $c = (c_1, \ldots, c_{N_{\text{RI}}})$ and normalize it, $\hat{c} = c/\|c\|$. Equation (18) says the divergent part of the matrix is $c\,c^{\mathrm{T}}$ times a number: it is a *rank-one* matrix, nonzero only along the single direction $\hat{c}$ of the auxiliary basis. We call that direction the *charge channel*. Any matrix can be split into its charge-channel part and the rest,

$$V = \underbrace{(\hat{c}^{\mathrm{T}} V \hat{c})\,\hat{c}\hat{c}^{\mathrm{T}}}_{\text{charge channel}} + \underbrace{\big(V - (\hat{c}^{\mathrm{T}} V \hat{c})\,\hat{c}\hat{c}^{\mathrm{T}}\big)}_{\text{rest}}, \tag{20}$$

and the whole problem of $\mathbf{q} = 0$ is that the number $\hat{c}^{\mathrm{T}} V \hat{c}$ is infinite. Two remarks matter later. First, not every auxiliary function is charged: an s-type Gaussian is, a p-type one is not, and a combination like $\phi_1 - (c_1/c_2)\phi_2$ is neutral. Second, the charge channel also contains a *finite* part — the interaction of a charged function with its own neighbourhood, the second term of Eq. (18) projected onto $\hat{c}$ — which is physical and must be kept. Only the divergent constant may go.

5. D_{box}: the constant that is removed

The divergent number is the lattice sum of $1/|\mathbf{R}|$ over the box. Replacing the sum by the integral over the region $\mathscr{P}$ that the summed cells cover — the parallelogram $\{u_1\mathbf{a}_1 + u_2\mathbf{a}_2 : |u_1|, |u_2| \le F\}$ with $F = 3^{s-1}/2$, one cell per lattice point — defines

$$D_{\text{box}} = \frac{1}{A}\int_{\mathscr{P}} \frac{\mathrm{d}^2 r}{|\mathbf{r}|}. \tag{21}$$

In polar coordinates $\mathrm{d}^2 r = r\,\mathrm{d}r\,\mathrm{d}\theta$, the $1/r$ cancels, and the radial integral is just the distance $r_{\max}(\theta)$ from the origin to the edge of the parallelogram in direction θ:

$$D_{\mathrm{box}} = \frac{1}{A}\int_0^{2\pi} \mathrm{d}\theta\; r_{\max}(\theta) = \frac{1}{A}\oint_{\partial\mathscr{P}} \frac{\mathbf{r}\times\mathrm{d}\mathbf{r}}{|\mathbf{r}|}, \tag{22}$$

a one-dimensional integral along the four edges of the box, done by Gauss–Legendre quadrature. It is a pure number fixed by the cell shape and the box size: 2.80 for the 6 Å water cell at $s = 3$, 1.52 for the h-BN cell at $s = 4$.

6. The Coulomb matrix used for W

Subtracting the constant along the charge direction, with the *unnormalized* charges so that the prefactor of Eq. (18) is matched,

$$V_{PQ}(0) = V^{\mathrm{lat}}_{PQ}(0) - D_{\mathrm{box}}\, c_P\, c_Q. \tag{23}$$

This removes exactly the first term of Eq. (18) and nothing else. What is left in the charge channel is small and finite: $\hat{c}^{\mathrm{T}} V \hat{c}$ goes from 796 to 8.9 Ha for the water cell and from 7.4×10^4 to -139 Ha for h-BN. In reciprocal space this is the familiar "omit the $\mathbf{G} = 0$ term" of the Coulomb sum; Eq. (23) is its real-space form.

What one must *not* do is project the charge channel out altogether, $V \to \Pi V \Pi$ with $\Pi = 1 - \hat{c}\hat{c}^{\mathrm{T}}$. That would also discard the finite part of the charge channel (the second remark above), which step 3 does not put back, and it was measured to inflate every Σ^c matrix element by a factor of 5–9 on the water test.

G. Step 2: ε and W from the finite $V(0)$

With the RI metric $M_{PQ} = (P|v_\omega|Q)$ and V from Eq. (23),

$$\varepsilon(\mathrm{i}\omega) = 1 + V^{1/2} M^{-1} \chi(\mathrm{i}\omega) M^{-1} V^{1/2}, \tag{24}$$

$$W^c(\mathrm{i}\omega) = V^{1/2}\big(\varepsilon^{-1}(\mathrm{i}\omega) - 1\big)V^{1/2}, \tag{25}$$

$$W^{\mathrm{MIC}}(\mathrm{i}\omega) = M^{-1}\, W^c(\mathrm{i}\omega)\, M^{-1}. \tag{26}$$

This is the only k-point, so W^{MIC} is Eq. (26) with weight one. (The code's two-mesh extrapolation must be switched off for this; with it, Γ would enter W with weight $1/(1-\sqrt{2})$ in some matrix elements and weight 0 in others.) W^{MIC} is transformed to imaginary time and brought to the grid,

$$W^{\mathrm{MIC}}_{\ell\ell'}(\mathrm{i}\tau) = \sum_{PQ} Z_{\ell P}\, W^{\mathrm{MIC}}_{PQ}(\mathrm{i}\tau)\, Z_{\ell' Q}. \tag{27}$$

H. Step 3: the charge channel, added back analytically

1. The constant function and its auxiliary charges

The charge channel is represented by one extra basis function, the normalized constant,

$$\phi_0(\mathbf{r}) = \frac{1}{\sqrt{A}}, \qquad V_{00}(\mathbf{q}) = \frac{2\pi}{q}. \tag{28}$$

$2\pi/q$ is the exact two-dimensional Coulomb head; it is finite for every $q > 0$ and the $1/q$ singularity is integrable in two dimensions, which is why integrating over the Brillouin zone works where sampling at $q = 0$ does not.

At a small but finite $\mathbf{q}$ the charge channel is the Bloch-modulated constant $\mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{r}}/\sqrt{A}$, and a fitted density $\rho = \sum_P \rho_P \phi_P$ couples to it through its Fourier component,

$$\begin{aligned}\langle\phi_{0,\mathbf{q}}|\rho\rangle = \frac{\hat{\rho}(\mathbf{q})}{\sqrt{A}} &= \frac{1}{\sqrt{A}}\sum_P \rho_P\, \hat{\phi}_P(\mathbf{q}) \\ &\simeq \frac{1}{\sqrt{A}}\sum_P \rho_P\, c_P\, \mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{R}_P}, \qquad c_P = \int \mathrm{d}\mathbf{r}\,\phi_P(\mathbf{r}),\end{aligned} \tag{29}$$

with $\mathbf{R}_P$ the centre of the auxiliary Gaussian ϕ_P. The last step is the point-charge limit $q_{\max}\sigma_P \ll 1$ of the Gaussian form factor; only s-type functions carry a charge. So the charge channel needs no fit of its own: the auxiliary charges c_P and the atomic positions are everything, and the correction is applied in the auxiliary basis (Sec. II H 5), reaching the grid through the same Z as every other part of W.

An earlier version gave ϕ_0 grid coefficients $Z_{\ell 0}$ from the fit of Eq. (8) with the constant as the target (solved globally by conjugate gradients, since its right-hand side does not decay) and applied the correction per grid pair, $Z_{\ell 0} Z_{\ell' 0} K_{\mathrm{h}}(\mathbf{R}_{\ell\ell'})$. That is numerically unusable. The RI-RS interpolation functions ξ_ℓ far from the atoms carry fit weights of 10^4–10^8 of alternating sign (on the 3×3 h-BN test cell, $\max|Z_{\ell 0}| = 2\times10^4$ at points with a density of 10^{-9}). They cancel exactly in linear functionals of a density, $\sum_\ell \rho(\mathbf{r}_\ell) Z_{\ell 0}$ gives $1/\sqrt{A}$ for a diagonal pair and 3×10^{-6} for a transition density, but not once each point is weighted by a different kernel value $K(\mathbf{R}_{\ell\ell'})$: the odd wing came out wrong by orders of magnitude and split symmetry-equivalent states by more than 1 eV, the even head survived only on large cells where the kernel is nearly flat over neighbouring points.

2. The neutral Gaussians

The Gaussians are made neutral by projecting out the charge direction,

$$\Pi = 1 - \hat{c}\hat{c}^{\mathrm{T}}, \qquad V_b = \Pi V^{\mathrm{lat}}(0)\,\Pi, \tag{30}$$

so that V_b is finite (neutral functions interact as R^{-3}). The two maps between a fitted grid density and the neutral basis are

$$F^{\mathrm{dens}} = M^{-1}\,\Pi V_b^{1/2}, \qquad F^{\mathrm{ext}} = M^{-1}\, V^{\mathrm{lat}}(0)\,\Pi V_b^{-1/2}, \tag{31}$$

the first for the density side of ε, the second for the potential side of W. We also write $G^{\mathrm{ext}} = MF^{\mathrm{ext}} = V^{\mathrm{lat}}(0)\,\Pi V_b^{-1/2}$ for the potential-side map in the representation of W^c.

3. *The response in the charge channel*

The constant function couples to the electrons through the transition dipoles. Under periodic boundary conditions the position operator is ill defined, so they are taken in the velocity gauge,

$$\mu^{\alpha}_{ai} = \frac{\langle a|[H, r^{\alpha}]|i\rangle}{\varepsilon_a - \varepsilon_i}, \qquad [H, r^{\alpha}] = -\partial_{\alpha} + [V_{\mathrm{nl}}, r^{\alpha}], \tag{32}$$

and checked against the Thomas–Reiche–Kuhn sum rule

$$\frac{2}{n_{\mathrm{spin}}} \sum_{\sigma} \sum_{ia} 2\,(\varepsilon_a - \varepsilon_i)\,|\mu^{\alpha}_{ai}|^2 = N_{\mathrm{el}}, \tag{33}$$

which a finite basis can only undershoot (h-BN: 0.64/0.64/0.41). The head and the wings of the polarizability in imaginary time are

$$Y_{\alpha\beta}(\mathrm{i}\tau) = \frac{2}{n_{\mathrm{spin}} A} \sum_{\sigma} \sum_{ia} \mu^{\alpha}_{ai} \mu^{\beta}_{ai}\, \mathrm{e}^{-(\varepsilon_a - \varepsilon_i)\tau}, \tag{34}$$

$$X_{\ell\alpha}(\mathrm{i}\tau) = \frac{2}{n_{\mathrm{spin}} \sqrt{A}} \sum_{\sigma} \sum_{ia} \psi_i(\mathbf{r}_{\ell})\, \psi_a(\mathbf{r}_{\ell})\, \mu^{\alpha}_{ai}\, \mathrm{e}^{-(\varepsilon_a - \varepsilon_i)\tau}, \tag{35}$$

and are cosine-transformed to imaginary frequency like χ. $Y(\mathrm{i}\omega \to 0)$ is the polarizability per area of the layer (h-BN: 1.35 bohr).

4. *Inverting ε in the extended basis*

With the constant as index 0, the dielectric matrix at small $\mathbf{q}$ is a 2×2 block matrix,

$$\varepsilon(\mathbf{q}, \mathrm{i}\omega) = \begin{pmatrix} h & w^{\dagger} \\ w & B \end{pmatrix}, \tag{36}$$

$$h = 1 + 2\pi q\, \hat{\mathbf{q}}^{\mathrm{T}} Y \hat{\mathbf{q}}, \qquad w_P = -\mathrm{i}\sqrt{2\pi q}\,(E\hat{\mathbf{q}})_P, \tag{37}$$

$$B(\mathrm{i}\omega) = 1 + F^{\mathrm{dens\,T}} \chi(\mathrm{i}\omega) F^{\mathrm{dens}}, \quad E(\mathrm{i}\omega) = F^{\mathrm{dens\,T}} Z^{\mathrm{T}} X(\mathrm{i}\omega). \tag{38}$$

B is the ordinary Gaussian dielectric matrix at Γ in the neutral basis; h and w are the new head and wings. B needs no new grid contraction: ZF^{dens} maps the grid to the neutral basis, so $F^{\mathrm{dens\,T}}(Z^{\mathrm{T}} \chi_{\ell\ell'} Z) F^{\mathrm{dens}}$ contains exactly the $\chi_{PQ}(\Gamma)$ of Eq. (13). Inverting the block matrix and keeping the head and the wings of $W^c = V^{1/2}(\varepsilon^{-1} - 1)V^{1/2}$ gives closed forms. All of the $\mathbf{q}$ dependence collapses into one scalar,

$$a(\hat{\mathbf{q}}, \mathrm{i}\omega) = \hat{\mathbf{q}}^{\mathrm{T}} S \hat{\mathbf{q}}, \qquad S = Y - E^{\mathrm{T}} B^{-1} E, \tag{39}$$

$$T_{P\alpha} = \left[V^{\mathrm{lat}}(0)\,\Pi V_b^{-1/2} B^{-1} E\right]_{P\alpha}, \tag{40}$$

(S is the 3×3 head with the body screening folded in, T the wing vector in the auxiliary basis — $MF^{\mathrm{ext}} B^{-1} E = G^{\mathrm{ext}} B^{-1} E$, i.e. in the same representation as W^c of Eq. (25) — both obtained by linear solves with B), and then

$$W^c_{\mathrm{h}}(\mathbf{q}, \mathrm{i}\omega) = -\frac{4\pi^2 a}{1 + 2\pi q a}, \tag{41}$$

$$W^c_{\mathrm{w},P}(\mathbf{q}, \mathrm{i}\omega) = \frac{\mathrm{i}\,2\pi \sum_{\alpha} \hat{q}_{\alpha} T_{P\alpha}}{1 + 2\pi q a}. \tag{42}$$

Both are finite at $q \to 0$: the head of ε^{-1} is proportional to q and cancels the $1/q$ of V_{00}, the wing is proportional to $\sqrt{q}$ and cancels $1/\sqrt{q}$. The head is a number, the wing is a vector over the auxiliary functions and linear in $\hat{\mathbf{q}}$, which is what allows the angular integral to be done once into a table. The explicit factors of i cancel in the Schur complement $h - w^{\dagger} B^{-1} w = 1 + 2\pi q a$ and reappear in W only as a quantity plus its own conjugate, so every stored array is real.

5. *Integral over the Brillouin zone and the correction to W*

The head and wings are now integrated over the whole small Brillouin zone of the supercell, with the phase $\mathrm{e}^{-\mathrm{i}\mathbf{q}\cdot\mathbf{R}}$ of Eq. (29) that turns $W(\mathbf{q})$ into a function of the separation $\mathbf{R} = \mathrm{MIC}(\mathbf{R}_P - \mathbf{R}_Q)$ of the atoms carrying two auxiliary functions. Because a is even and the wing odd in $\hat{\mathbf{q}}$, only cos survives in the head and only sin in the wing, and both results are real:

$$K_{\mathrm{h}}(\mathbf{R}, \mathrm{i}\omega) = \frac{A}{(2\pi)^2} \int_{\mathrm{BZ}} \mathrm{d}^2 q \left[-\frac{4\pi^2 a}{1 + 2\pi q a} \right] \cos(\mathbf{q}\cdot\mathbf{R}), \tag{43}$$

$$K_{\mathrm{w},\alpha}(\mathbf{R}, \mathrm{i}\omega) = \frac{A}{(2\pi)^2} \int_{\mathrm{BZ}} \mathrm{d}^2 q \left[\frac{2\pi \hat{q}_{\alpha}}{1 + 2\pi q a} \right] \sin(\mathbf{q}\cdot\mathbf{R}). \tag{44}$$

The prefactor $A/(2\pi)^2$ makes this the Brillouin-zone *average*, the continuum limit of $\frac{1}{N_k}\sum_k$. The integrals are done in polar coordinates (24 Gauss–Legendre directions per zone edge, 32 radial points), tabulated on a 128×128 mesh of $\mathbf{R}$ over the minimum-image cell and interpolated bilinearly at the atom-pair separations. The correction to the screened interaction in the auxiliary basis is

$$\Delta W_{PQ}(\mathrm{i}\omega) = \frac{c_P c_Q}{A} K_{\mathrm{h}}(\mathbf{R}_{PQ}, \mathrm{i}\omega) + \frac{1}{\sqrt{A}} \sum_{\alpha} \left[T_{P\alpha} c_Q - c_P T_{Q\alpha} \right] K_{\mathrm{w},\alpha}(\mathbf{R}_{PQ}, \mathrm{i}\omega), \tag{45}$$

symmetric in $P \leftrightarrow Q$ (K_{h} is even and K_{w} odd in $\mathbf{R}$, the wing bracket antisymmetric). It is transformed to imaginary time with the cosine weights of the minimax grid and added to $W^c_{PQ}(\mathrm{i}\tau)$ *before* Eq. (26), so that it reaches the grid pairs through M^{-1} and Z exactly like the regular W. Its head term is a uniform potential $K_{\mathrm{h}}(0)/A$ in real space (-0.29 eV for h-BN). The cost is negligible: one table lookup per auxiliary pair and time point.

I. Exchange self-energy

The exchange part uses the truncated Coulomb operator v_{tc} with a cutoff radius equal to half the minimum cell dimension (15.2 Å for the h-BN cell), which is finite without any treatment of the charge channel and is unaffected by everything above:

$$P_{\mu\nu} = \sum_{i}^{\mathrm{occ}} C_{\mu i} C_{\nu i}, \qquad P_{\ell\ell'} = \sum_{\mu\nu} \Phi_\mu(\mathbf{r}_\ell) P_{\mu\nu} \Phi_\nu(\mathbf{r}_{\ell'}), \tag{46}$$

$$V^{\mathrm{tc}}_{\ell\ell'} = \sum_{PQ} Z_{\ell P} \left[M^{-1} V^{\mathrm{tc}} M^{-1} \right]_{PQ} Z_{\ell' Q}, \tag{47}$$

$$\Sigma^{x}_{\lambda\sigma} = -\sum_{\ell\ell'} \Phi_\lambda(\mathbf{r}_\ell)\, P_{\ell\ell'}\, V^{\mathrm{tc}}_{\ell\ell'}\, \Phi_\sigma(\mathbf{r}_{\ell'}). \tag{48}$$

J. Correlation self-energy and quasiparticle energies

The correlation self-energy is an element-wise product on the grid of the Green's function with the *full* screened interaction, the regular part of Eq. (27) plus the charge-channel correction of Eq. (45), in the two time branches:

$$\Sigma^{c}_{\ell\ell'}(-\mathrm{i}|\tau|) = -\,G^{\mathrm{occ}}_{\ell\ell'}(\mathrm{i}|\tau|) \left[W^{\mathrm{MIC}}_{\ell\ell'}(\mathrm{i}\tau) + \Delta W_{\ell\ell'}(\mathrm{i}\tau) \right], \tag{49}$$

$$\Sigma^{c}_{\ell\ell'}(+\mathrm{i}|\tau|) = +\,G^{\mathrm{vir}}_{\ell\ell'}(\mathrm{i}|\tau|) \left[W^{\mathrm{MIC}}_{\ell\ell'}(\mathrm{i}\tau) + \Delta W_{\ell\ell'}(\mathrm{i}\tau) \right], \tag{50}$$

$$\Sigma^{c}_{\lambda\sigma}(\pm\mathrm{i}|\tau|) = \sum_{\ell\ell'} \Phi_\lambda(\mathbf{r}_\ell)\, \Sigma^{c}_{\ell\ell'}(\pm\mathrm{i}|\tau|)\, \Phi_\sigma(\mathbf{r}_{\ell'}). \tag{51}$$

In the implementation $W^{\mathrm{MIC}}_{\ell\ell'}$ is built panel by panel as $Z W^{\mathrm{MIC}}_{PQ} Z^{\mathrm{T}}$, with ΔW already inside W^{MIC}_{PQ} (Eq. (45)), and the two Hadamard products with G^{occ} and G^{vir} share that panel. The panels are restricted to the pairs within the geometric cutoff of the Σ^c panels, which is exact because the Hadamard product with the short-ranged G vanishes outside it.

The self-energy is projected onto the Kohn–Sham states, $\Sigma^{c}_{nn}(\pm\mathrm{i}|\tau|) = \sum_{\lambda\sigma} C_{\lambda n} \Sigma^{c}_{\lambda\sigma}(\pm\mathrm{i}|\tau|) C_{\sigma n}$, split into its even and odd parts in τ and transformed to imaginary frequency with the cosine and sine weights of Eq. (3),

$$\Sigma^{c}_{nn}(\mathrm{i}\omega_k) = \sum_j \gamma^{\cos}_{kj} \frac{\Sigma^{c}_{nn}(\mathrm{i}\tau_j) + \Sigma^{c}_{nn}(-\mathrm{i}\tau_j)}{2} + \sum_j \gamma^{\sin}_{kj} \frac{\Sigma^{c}_{nn}(\mathrm{i}\tau_j) - \Sigma^{c}_{nn}(-\mathrm{i}\tau_j)}{2}. \tag{52}$$

$\Sigma^{c}_{nn}(\mathrm{i}\omega)$ is continued to the real axis by a Padé approximant fitted to the imaginary-frequency points, and the quasiparticle energies follow from

$$\varepsilon^{GW}_n = \varepsilon_n + \mathrm{Re}\,\Sigma^{c}_{nn}(\varepsilon^{GW}_n) + \Sigma^{x}_{nn} - v^{xc}_{nn}, \tag{53}$$

solved by a Newton search for the crossing of the quasiparticle equation. For G_0W_0 this is done once with G built from the Kohn–Sham eigenvalues; for evGW_0 the Green's function is rebuilt from the quasiparticle energies and Eqs. (11)–(52) are iterated to self-consistency with W frozen.

III. THEORY FOR THREE-DIMENSIONAL BULK CELLS

A. Substitutions

The same construction applies to a bulk cell (`PERIODIC XYZ`) with three substitutions. The measure is the cell volume $\Omega = |\det(\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3)|$, $A \to \Omega$ in Eqs. (29), (34), (35) and (45), the reciprocal vectors are the three $\mathbf{b}_i$ with $\mathbf{b}_i \cdot \mathbf{a}_j = 2\pi\delta_{ij}$, and the charge mode is $\phi_0 = \mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{r}}/\sqrt{\Omega}$ with $v_{00} = 4\pi/q^2$. Everything else — the dipoles, the response Y, X, B, E, the block inverse and the transform $\mathrm{i}\omega \to \mathrm{i}\tau$ — is shared with the 2D case.

B. The Coulomb matrix: Ewald sum without $\mathbf{G} = 0$

The finite box sum of Eq. (14) cannot be used in three dimensions, not even after the subtraction of Eq. (23): between two *neutral* (dipolar) functions the sum of $1/r$ over a 3D box is conditionally convergent, with a surface term $\frac{1}{2} d_\alpha d_\beta \int_{\mathrm{box}} \partial_\alpha \partial_\beta r^{-1}$ that depends on the shape of the box and does not vanish as the box grows (the Lorentz depolarization of a finite sample). The matrix that the head-and-wing model assumes is the tin-foil limit, the periodic Coulomb matrix with the $\mathbf{G} = 0$ Fourier coefficient of $1/r$ omitted, which is obtained as an Ewald sum,

$$V_{PQ}(0) = \sum_{\mathbf{L} \in \mathrm{box}} \left(P\,0 \,\middle|\, \frac{\mathrm{erfc}(\omega r)}{r} \,\middle|\, Q\,\mathbf{L} \right) + \frac{4\pi}{\Omega} \sum_{\mathbf{G} \neq 0} \frac{\mathrm{e}^{-G^2/4\omega^2}}{G^2} \rho_P(\mathbf{G}) \rho_Q(\mathbf{G})^* - \frac{\pi}{\Omega\,\omega^2}\, c_P c_Q, \qquad \rho_P(\mathbf{G}) = \int \mathrm{d}\mathbf{r}\, \phi_P(\mathbf{r})\, \mathrm{e}^{\mathrm{i}\mathbf{G}\cdot\mathbf{r}}, \tag{54}$$

where $1/r = \mathrm{erfc}(\omega r)/r + \mathrm{erf}(\omega r)/r$, the erf part has the Fourier transform $(4\pi/G^2)\,\mathrm{e}^{-G^2/4\omega^2} = 4\pi/G^2 - \pi/\omega^2 + O(G^2)$, and dropping its $4\pi/G^2$ singularity leaves $-\pi/\omega^2$ at $\mathbf{G} = 0$. (Here ω is the Ewald splitting parameter, not an imaginary frequency.) The result is independent of ω, which checks both halves against each other; the code sets $\omega r_c = 8$ with $r_c = \frac{1}{2}(N-1)h_{\min}$ the coverage radius of the image box ($N = 3^{s-1}$ images per direction, $h_{\min}$ the smallest cell height $2\pi/|\mathbf{b}_i|$), so that the erfc tail is negligible for every pair the box contains ($\mathrm{erfc}(8) < 10^{-28}$; the Gaussian ex-

tents of the auxiliary functions use a few bohr of that margin). $\rho_P(\mathbf{G})$ is analytic for Gaussians (one-dimensional moments $m_l(p) = \int \mathrm{d}x\, x^l \mathrm{e}^{-\alpha x^2}\mathrm{e}^{\mathrm{i}px}$ by the recursion $m_l = (\mathrm{i}p\, m_{l-1} + (l-1)m_{l-2})/2\alpha$). This $V(0)$ is finite and positive on the charge direction, so no projector is needed: $\Pi = 1$, $V_b = V(0)$, $F^{\mathrm{dens}} = M^{-1}V^{1/2}$ and $G^{\mathrm{ext}} = V^{1/2}$, so that Eq. (40) becomes

$$T = V(0)\,V^{-1/2}B^{-1}E = V^{1/2}B^{-1}E. \tag{55}$$

The regular path of Sec. II G uses the same matrix with nothing subtracted (the same choice as the 3D branch of CP2K #6447 [2]).

C. Head and wings

In the extended basis the head is $h = 1 + 4\pi\,\hat{\mathbf{q}}^{\mathrm{T}}Y\hat{\mathbf{q}}$ and the wing $w_P = -\mathrm{i}\sqrt{4\pi}\,(E\hat{\mathbf{q}})_P$, both independent of q: the q^2 of the dipole response cancels the $1/q^2$ of v_{00} where in two dimensions $2\pi q\,a$ was left. The Schur complement of the symmetrized dielectric matrix is

$$\tilde{\varepsilon}(\hat{\mathbf{q}},\mathrm{i}\omega) = 1 + 4\pi a(\hat{\mathbf{q}},\mathrm{i}\omega), \qquad a = \hat{\mathbf{q}}^{\mathrm{T}}S\hat{\mathbf{q}}, \tag{56}$$

with S from Eq. (39), and the block inverse gives

$$W^c_{\mathrm{h}}(\mathbf{q},\mathrm{i}\omega) = \frac{4\pi}{q^2}\Big(\frac{1}{1+4\pi a} - 1\Big) = -\frac{4\pi}{q^2}\,\frac{4\pi a}{1+4\pi a}, \tag{57}$$

$$W^c_{\mathrm{w},P}(\mathbf{q},\mathrm{i}\omega) = \frac{\mathrm{i}\,(4\pi/q)\sum_\alpha \hat{q}_\alpha T_{P\alpha}}{1+4\pi a}. \tag{58}$$

$1 + 4\pi a(\hat{\mathbf{q}},\mathrm{i}\omega)$ is the macroscopic dielectric constant of the cell along $\hat{\mathbf{q}}$. It must be positive, a being a quadratic form of the positive head Y minus the body correction; the code aborts if it is not.

D. Cell integrals

The Brillouin-zone average now carries $\Omega/(2\pi)^3$ and the volume element $q^2\mathrm{d}q\,\mathrm{d}\Omega_{\hat{\mathbf{q}}}$, which cancels the $1/q^2$ of the head, so that the radial integral is elementary:

$$\begin{aligned} K_{\mathrm{h}}(\mathbf{R},\mathrm{i}\omega) &= -\frac{\Omega}{(2\pi)^3}\int \mathrm{d}\Omega_{\hat{\mathbf{q}}}\,\frac{16\pi^2 a}{1+4\pi a}\int_0^{L(\hat{\mathbf{q}})}\mathrm{d}q\,\cos(q\,\hat{\mathbf{q}}\cdot\mathbf{R}) \\ &= -\frac{\Omega}{(2\pi)^3}\int \mathrm{d}\Omega_{\hat{\mathbf{q}}}\,\frac{16\pi^2 a}{1+4\pi a}\,\frac{\sin(Lt)}{t}, \end{aligned} \tag{59}$$

with $t = \hat{\mathbf{q}}\cdot\mathbf{R}$ and $L(\hat{\mathbf{q}})$ the distance from Γ to the boundary of the cell $\{\sum_i s_i\mathbf{b}_i/2,\ |s_i| \le 1\}$ along $\hat{\mathbf{q}}$. In the wing the volume element combines with the $1/q$ of Eq. (58),

$$\begin{aligned} K_{\mathrm{w},\alpha}(\mathbf{R},\mathrm{i}\omega) &= \frac{\Omega}{(2\pi)^3}\int \mathrm{d}\Omega_{\hat{\mathbf{q}}}\,\frac{4\pi\,\hat{q}_\alpha}{1+4\pi a}\int_0^{L(\hat{\mathbf{q}})}\mathrm{d}q\,q\,\sin(q\,\hat{\mathbf{q}}\cdot\mathbf{R}) \\ &= \frac{\Omega}{(2\pi)^3}\int \mathrm{d}\Omega_{\hat{\mathbf{q}}}\,\frac{4\pi\,\hat{q}_\alpha}{1+4\pi a}\,\frac{\sin(Lt) - Lt\cos(Lt)}{t^2}. \end{aligned} \tag{60}$$

Both radial results are Taylor expanded below $|Lt| = 0.02$, where the closed forms cancel,

$$\begin{aligned} \frac{\sin(Lt)}{t} &= L\Big(1 - \frac{(Lt)^2}{6} + \frac{(Lt)^4}{120} - \cdots\Big), \\ \frac{\sin(Lt) - Lt\cos(Lt)}{t^2} &= L^2\Big(\frac{Lt}{3} - \frac{(Lt)^3}{30} + \frac{(Lt)^5}{840} - \cdots\Big), \end{aligned} \tag{61}$$

with a next omitted term of relative size 10^{-13}.

The solid-angle integral is a cubature over the six faces of that cell. Face pair i sits at $s_i = \pm 1$ and is parametrized by (s_j, s_k), a point on it is $\mathbf{q} = \pm\mathbf{b}_i/2 + s_j\mathbf{b}_j/2 + s_k\mathbf{b}_k/2$, and its solid-angle element seen from Γ is

$$\mathrm{d}\Omega = \frac{|\mathbf{q}\cdot\mathrm{d}\mathbf{S}|}{q^3} = \frac{|\mathbf{q}\cdot(\mathbf{b}_j\times\mathbf{b}_k)|}{4\,q^3}\,\mathrm{d}s_j\,\mathrm{d}s_k, \tag{62}$$

integrated with an 8×8 Gauss–Legendre rule per face (6×64 directions $\hat{\mathbf{q}}_d$ with weights ϖ_d), so that $\int_{\mathrm{cell}}\mathrm{d}^3q\,f(\hat{\mathbf{q}})g(q) = \sum_d \varpi_d \int_0^{L(\hat{\mathbf{q}}_d)}\mathrm{d}q\,q^2 f(\hat{\mathbf{q}}_d)g(q)$. Both signs of every face are included, which makes the direction set symmetric under $\hat{\mathbf{q}} \to -\hat{\mathbf{q}}$ and the cos/sin split of the kernels exact. The rule is checked by $\sum_d \varpi_d L_d^3/3 = (2\pi)^3/\Omega$.

E. Where the bulk correction is applied: in the orbital basis, not through the RI fit

In two dimensions the head is a ~ 30 meV effect and Eq. (45) can be applied in the auxiliary basis. In three dimensions the head potential $K_{\mathrm{h}}(0,\mathrm{i}\omega)/\Omega$ is of order eV (-3.5 eV for a $3\times3\times3$ diamond cell with $\varepsilon_\infty = 3$) and scales as $1/L$, and the same application fails. $c_P c_Q K_{\mathrm{h}}(\mathbf{R}_{PQ})$ is the projection of a smooth, long-wavelength kernel onto atom-centred Gaussians; its fit through $M^{-1}\cdot M^{-1}$ in Eq. (26) demands large cancelling coefficients, and the neglected form factors break the cancellation. On the 54-atom diamond cell the head alone, applied in the RI basis, gives a gap of -96 eV and splits a degenerate triplet by 0.4 eV — which a uniform potential cannot do. The bulk correction is therefore added where it is an exact functional of the orbital-pair densities: to Σ^c in the orbital basis, after Eq. (51), with $\Delta W_{\ell\ell'} = 0$ in Eqs. (49)–(50). No RI fit of a kernel is involved anywhere.

F. The orbital-pair kernel

Between orbital-pair densities $\rho_{nm} = \psi_n\psi_m$ the cell average of the correction needs only the coupling of the pair to the charge mode,

$$\langle mn|\phi_0\rangle = \frac{1}{\sqrt{\Omega}}\int \mathrm{d}\mathbf{r}\,\rho_{mn}(\mathbf{r})\,\mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{r}} = \frac{\delta_{mn} + \mathrm{i}\,\mathbf{q}\cdot\mathbf{d}_{mn}}{\sqrt{\Omega}} + O(q^2), \tag{63}$$

with the velocity-gauge dipole between *any* two orbitals,

$$\mathbf{d}_{nm} = \langle n|\mathbf{r}|m\rangle = \frac{\langle n|[H,\mathbf{r}]|m\rangle}{\varepsilon_n - \varepsilon_m}, \tag{64}$$

which vanishes between folded states of different crystal momentum, where the commutator vanishes, and is set to zero inside degenerate manifolds ($|\varepsilon_n - \varepsilon_m| < 10^{-5}$ Ha), where it is not defined. Collecting $\langle nm|\Delta W(\mathbf{q})|mn\rangle$ from the head (57) and the wings (58) and averaging over the cell gives the real kernel

$$\begin{aligned}\hat{W}_{nm}(\mathrm{i}\omega) = w(\mathrm{i}\omega)\,\delta_{nm} + \sum_{\alpha\beta} B^{\mathrm{hd}}_{\alpha\beta}(\mathrm{i}\omega)\, d_{nm,\alpha} d_{nm,\beta} \\ + 2\sqrt{\Omega}\sum_{\alpha\beta} A^{\mathrm{w}}_{\alpha\beta}(\mathrm{i}\omega)\, V_{nm,\alpha}(\mathrm{i}\omega)\, d_{nm,\beta},\end{aligned} \tag{65}$$

with the three ingredients

$$w(\mathrm{i}\omega) = \frac{K_{\mathrm{h}}(\mathbf{0},\mathrm{i}\omega)}{\Omega}, \tag{66}$$

$$A^{\mathrm{w}}_{\alpha\beta}(\mathrm{i}\omega) = \frac{1}{(2\pi)^3}\int_{\mathrm{cell}} \mathrm{d}^3q\, \frac{4\pi}{\tilde{\varepsilon}(\hat{\mathbf{q}},\mathrm{i}\omega)}\,\hat{q}_\alpha\hat{q}_\beta, \tag{67}$$

$$B^{\mathrm{hd}}_{\alpha\beta}(\mathrm{i}\omega) = \frac{1}{(2\pi)^3}\int_{\mathrm{cell}} \mathrm{d}^3q\, 4\pi\Big(\frac{1}{\tilde{\varepsilon}(\hat{\mathbf{q}},\mathrm{i}\omega)} - 1\Big)\hat{q}_\alpha\hat{q}_\beta, \tag{68}$$

evaluated with the face cubature as $\int_{\mathrm{cell}} \mathrm{d}^3q\, f(\hat{\mathbf{q}}) = \sum_d \varpi_d L_d^3 f(\hat{\mathbf{q}}_d)/3$, and the wing potential

$$\begin{aligned}V_{nm,\alpha}(\mathrm{i}\omega) &= \int \mathrm{d}\mathbf{r}\,\rho_{nm}(\mathbf{r})\, w_\alpha(\mathbf{r};\mathrm{i}\omega) \\ &= \sum_\ell \rho_{nm}(\mathbf{r}_\ell)\,\big[ZM^{-1}T(\mathrm{i}\omega)\big]_{\ell\alpha}, \\ w_\alpha &= \sum_P \hat{\phi}_P\, T_{P\alpha},\end{aligned} \tag{69}$$

where $\hat{\phi}_P$ denotes the RI-RS dual of ϕ_P, i.e. the functional $\int \rho\,\hat{\phi}_P = \sum_\ell \rho(\mathbf{r}_\ell)(ZM^{-1})_{\ell P}$ that maps a density to its coefficient in the representation of W^{MIC}. With $\tilde{w} = ZM^{-1}T$ this is evaluated as

$$V_{nm,\alpha} = \big[C^{\mathrm{T}}\big(\Phi^{\mathrm{T}}\,\mathrm{diag}(\tilde{w}_{\cdot\alpha})\,\Phi\big)C\big]_{nm}. \tag{70}$$

The head is a uniform potential (order $1/L$), the head-dipole term is the $O(q^2)$ remainder of the head, and the wing couples the dipole of the pair density to the potential w_α that the uniform field induces in the body ($A^{\mathrm{w}}, B^{\mathrm{hd}} \propto 1/\Omega$ and $T \propto 1/\sqrt{\Omega}$, so both are of order $1/\Omega$). The wing carries a single $1/\sqrt{\Omega}$ from Eq. (63) against the $\Omega/(2\pi)^3$ of the average, hence $2\sqrt{\Omega}$; the head terms carry $|\langle mn|\phi_0\rangle|^2 \propto 1/\Omega$, which the average cancels. V_{nm} is a linear functional of ρ_{nm} and is exact on the RI-RS grid, unlike the pair kernel $c_P c_Q K_{\mathrm{h}}(\mathbf{R}_{PQ})$ of the RI-space application, Eq. (45).

G. Correction of the correlation self-energy

$\hat{W}_{nm}$ is transformed to imaginary time with the cosine weights of the minimax grid, $\hat{W}_{nm}(\mathrm{i}\tau_j) = \sum_k \tilde{\gamma}^{\cos}_{jk}\hat{W}_{nm}(\mathrm{i}\omega_k)$. With the Σ^c construction of Eqs. (49)–(50) ($\Sigma^c(-\mathrm{i}|\tau|) = -G^{\mathrm{occ}}\circ W$, $\Sigma^c(+\mathrm{i}|\tau|) = +G^{\mathrm{vir}}\circ W$) the orbital-diagonal correction is

$$\Delta\Sigma^c_n(-\mathrm{i}|\tau|) = -\sum_m^{\mathrm{occ}} \mathrm{e}^{-|\tau|(\varepsilon_F-\varepsilon_m)}\,\hat{W}_{nm}(\mathrm{i}\tau), \tag{71}$$

$$\Delta\Sigma^c_n(+\mathrm{i}|\tau|) = +\sum_m^{\mathrm{vir}} \mathrm{e}^{-|\tau|(\varepsilon_m-\varepsilon_F)}\,\hat{W}_{nm}(\mathrm{i}\tau), \tag{72}$$

added to the AO matrices of Eq. (51) as

$$\Delta\Sigma^c_{\mu\nu}(\pm\mathrm{i}|\tau|) = \Big[(\mathsf{S}C)\,\mathrm{diag}\big(\Delta\Sigma^c_n(\pm\mathrm{i}|\tau|)\big)\,(\mathsf{S}C)^{\mathrm{T}}\Big]_{\mu\nu}, \tag{73}$$

with S the AO overlap matrix. Because $C^{\mathrm{T}}\mathsf{S}C = 1$ this folds onto the band and density-of-states k-points exactly like the regular Σ^c. For the head alone, $\hat{W}_{nm} = w\,\delta_{nm}$, Eqs. (71)–(73) reduce to

$$\begin{aligned}\Delta\Sigma^c_{\mu\nu}(-\mathrm{i}|\tau|) &= -w(\mathrm{i}\tau)\,\big[\mathsf{S}C_{\mathrm{occ}}\,\mathrm{e}^{-|\tau|(\varepsilon_F-\varepsilon)}C^{\mathrm{T}}_{\mathrm{occ}}\mathsf{S}\big]_{\mu\nu}, \\ \Delta\Sigma^c_{\mu\nu}(+\mathrm{i}|\tau|) &= +w(\mathrm{i}\tau)\,\big[\mathsf{S}C_{\mathrm{vir}}\,\mathrm{e}^{-|\tau|(\varepsilon-\varepsilon_F)}C^{\mathrm{T}}_{\mathrm{vir}}\mathsf{S}\big]_{\mu\nu},\end{aligned} \tag{74}$$

whose orbital diagonal is $\mp w(\mathrm{i}\tau)\,\mathrm{e}^{-|\tau(\varepsilon_n-\varepsilon_F)|}$ for occupied and empty n: the $\pm w/2$ shift of the COHSEX head in the static limit. A uniform potential couples a state only to itself, $\langle m|\mathrm{e}^{\mathrm{i}\mathbf{q}\cdot\mathbf{r}}|n\rangle = \delta_{mn} + O(q)$, and therefore cannot split a degenerate manifold; the head-dipole and wing terms carry the $O(\mathbf{q}\cdot\mathbf{d}_{mn})$ corrections of the same expansion.

IV. IMPLEMENTATION IN CP2K

A. Activation

The correction lives in the module `gw_gamma_correction` and is switched on automatically for the large-cell Γ drivers, the RI-RS one and the exact-RI tensor one (`gw_tensor_large_cell_gamma`), when (i) the system is a slab (`PERIODIC XY`, head $2\pi/q$) or a bulk cell (`PERIODIC XYZ`, head $4\pi/q^2$), and (ii) the mesh for χ, ε, W is `KPOINTS_W 1 1 1`. Only the $1\times1\times1$ mesh samples $\mathbf{q} = 0$: the setup keeps $n_1n_2n_3/2$ points of an $n_1\times n_2\times n_3$ mesh, one per $\pm\mathbf{k}$ pair, and for odd $n > 1$ the integer division drops the self-paired Γ point. For this case the extrapolation mesh is dropped (otherwise Γ would enter the s–s elements of W with the extrapolation weight -2.414 and every other element with weight 0). Wires and slabs whose vacuum direction is not z are not implemented. There is no input keyword. In the tensor driver only the optical response $S_{\alpha\beta}$, $T_{P\alpha}$ of Sec. II H is evaluated on the RI-RS grid, which is built for that purpose; χ, the body of W and Σ^c remain exact-RI tensor contractions. This gives a reference in which the RI-RS fit enters only through the correction, and it is used in Sec. V to separate fit errors from the Γ-point model.

B. Call sequence

Table I lists the routines in the order in which the *GW* driver calls them. `gamma_corr_init` runs before

the $Z_{\ell P}$ fit and sets only geometric quantities (layer area or cell volume, reciprocal vectors, metric head). The Coulomb side, the response and the cell integrals are built in step 4b of the Γ driver (`gw_ri_rs_large_cell_gamma` or `gw_tensor_large_cell_gamma`), after $\chi_{PQ}(\Gamma,\mathrm{i}\tau)$ exists and before `get_W_MIC` consumes it. The $V(0)$ of `gamma_corr_coulomb_at_zero` is kept distributed and reused by the regular W path instead of repeating the lattice sum. In 2D, `gamma_corr_add_to_W` adds $\Delta W_{PQ}(\mathrm{i}\tau)$ inside `compute_W_MIC`, before the contraction $M^{-1}W^{\mathrm{MIC}}M^{-1}$; in 3D it returns immediately and `gamma_corr_add_to_sigma` adds Eqs. (71)–(73) after `compute_Sigma_c` and before the quasiparticle solve.

The wing quantity $X_{\ell\alpha}$ of Eq. (35) is evaluated as the diagonal of $\Phi P^{\alpha}(\mathrm{i}\tau)\Phi^{\mathrm{T}}$ with the dense AO matrix $P^{\alpha} = C_{\mathrm{occ}}\,\mathrm{e}\,\mu^{\alpha}\,\mathrm{e}\,C_{\mathrm{vir}}^{\mathrm{T}}$, using that the exponential factorizes, $\mathrm{e}^{-(\varepsilon_a-\varepsilon_i)\tau} = \mathrm{e}^{-|\varepsilon_a-\varepsilon_F|\tau}\mathrm{e}^{-|\varepsilon_i-\varepsilon_F|\tau}$; this is formed by two ScaLAPACK multiplications and one sparse product on the block pattern of Φ, and is correct under any distribution of the MO coefficients.

C. Replicated or distributed dense algebra

The Coulomb maps and the response involve dense $N_{\mathrm{RI}}\times N_{\mathrm{RI}}$ algebra (V, Π, V_b, $V_b^{\pm1/2}$, M^{-1}, F^{dens}, G^{ext}, B and its inverse). The replicated path holds up to ten such arrays per rank. `gamma_corr_init` estimates that peak as $10\times 8N_{\mathrm{RI}}^2$ bytes, measures the memory available per MPI rank, and takes the distributed (ScaLAPACK) path when the estimate exceeds a quarter of it; the remaining three quarters are left for the distributed matrices of the regular W path. With 12 auxiliary functions per atom the replicated peak is 4 GB per rank at about 600 atoms and 46 GB at 2000 atoms. In the distributed path Π is filled locally, $V_b = \Pi V \Pi$, F^{dens} and G^{ext} are parallel matrix products, $V_b^{\pm1/2}$ come from one distributed eigendecomposition with the same relative eigenvalue cut, and per frequency B is inverted by a Cholesky factorization, $x = B^{-1}E$, $S -= E^{\mathrm{T}}x$, $T = G^{\mathrm{ext}}x$. Only $N_{\mathrm{RI}}\times 3$ arrays are replicated. The choice is automatic; the two paths agree to all printed digits on one rank. The remaining replicated arrays, the dipoles μ ($3n_{\mathrm{occ}}n_{\mathrm{vir}}$) and $X(\mathrm{i}\tau)$ on the grid ($3N_{\mathrm{grid}}N_\tau$), grow linearly with system size.

D. Cost

In 2D the correction costs one table lookup per auxiliary pair and time point in `gamma_corr_add_to_W`; the tables are built once from the 3×3 tensors S on a 128×128 mesh. In 3D the cell integrals are per-direction coefficients ($6\times8\times8$ directions) and the radial integrals are analytic. The orbital-pair kernel costs, per frequency, three products $\Phi^{\mathrm{T}}\mathrm{diag}(\tilde{w}_{\cdot\alpha})\Phi$ on the grid, six $n_{\mathrm{AO}}^2 n_{\mathrm{MO}}$ transforms to the MO basis, and an $n_{\mathrm{MO}}^2/n_{\mathrm{ranks}}$ element loop; the dipoles $\mathbf{d}_{nm}$ need three $n_{\mathrm{AO}}^2 n_{\mathrm{MO}}$ transforms once per spin.

E. Order of evaluation

1. $\Phi_\mu(\mathbf{r}_\ell)$, Eq. (4); $Z_{\ell P}$ sphere by sphere, Eqs. (5)–(9).
2. $G^{\mathrm{occ}}, G^{\mathrm{vir}}$, $\chi_{\ell\ell'}$, $\chi_{PQ}(\mathrm{i}\tau)\to\chi_{PQ}(\mathrm{i}\omega)$, Eqs. (11)–(13).
3. c_P, Π, V_b, F^{dens}, F^{ext}, Eqs. (16)–(31); dipoles and sum rule, Eqs. (32)–(33); Y, X, B, E, S, T, Eqs. (34)–(40); 2D: tables K_{h}, K_{w}, Eqs. (43)–(44); 3D: direction coefficients, w, A^{w}, B^{hd}, Eqs. (59), (66)–(68).
4. 2D: $V^{\mathrm{lat}}(0)\to V(0)$, Eqs. (14)–(23); 3D: the Ewald $V(0)$ of Eq. (54), nothing subtracted. ε, W^c, $W^c(\mathrm{i}\omega)\to W^c(\mathrm{i}\tau)$; 2D: $W^c_{PQ}(\mathrm{i}\tau)$ += $\Delta W_{PQ}(\mathrm{i}\tau)$, Eq. (45); then W^{MIC}, Eqs. (24)–(26).
5. Σ^x, Eq. (48).
6. Per time point and panel: $W^{\mathrm{MIC}}_{\ell\ell'}$, Eq. (27); Σ^c in both branches, Eqs. (49)–(51).
7. 3D: $\hat{W}_{nm}$, Eqs. (64)–(70); $\Delta\Sigma^c$, Eqs. (71)–(73).
8. Eqs. (52)–(53).

V. RESULTS

All reference calculations use the full-k-point RI-RS *GW* of J. Wilhelm, cp2k PR #6447, in the form in which it was merged into the CP2K master branch (commit `7b88fc63f1`). The Γ-only code was rebased onto the same master, so that every number of this section, ours and the reference, comes from one binary. That master also carries a change of the analytic continuation (PR #6474: the quasiparticle energy is the average over several central Padé orders instead of a single 16-parameter fit), which shifts every G_0W_0 gap of this section by +0.2 to +0.4 eV relative to our earlier standard-Padé numbers, identically in both codes, and leaves Hartree–Fock energies untouched; the differences between the two codes are unchanged within 6 meV. Both codes use the same orbital and auxiliary bases, pseudopotentials, cutoffs, minimax grid and RI settings. The PR requires its own optimised RI-RS grid (`RS_AO_RATIO` 7, about 90 points per atom with a DZVP basis); a three times denser grid changes the h-BN gap by 2 meV. Our runs use the tabulated atom-centred grids of CP2K (`data/ri_rs_grid`, about 340 points per atom). Odd meshes of the PR abort in its own Γ treatment, so its series are 12 to 36 in steps of 6 (2D) and 4 to 12 in steps of 2 (3D).

A. Validation of the slab correction: h-BN with the aug-SZV basis

Table II compares the Γ-only 8×8 h-BN supercell (128 atoms, aug-SZV-MOLOPT-GTH tier-1 orbital and RI bases, 15 Å vacuum, cutoff 600 Ry, 10 minimax points) with the PR at increasing meshes, at the primitive point $P = (11/32, 11/32)$ nearest to K that the supercell samples, and state by state at the point Q that the supercell folds onto its

TABLE I. Call sequence of the Γ-point correction in CP2K. "2D"/"3D" mark dimension-specific branches.

caller	routine	computes
`ri_rs_setup`	`gamma_corr_init`	activation; A or Ω, $\mathbf{b}_i$, metric head; dense-algebra mode
Γ driver, step 4b	`gamma_corr_build_coulomb_maps`	c, Π, V_b, F^{dens}, G^{ext}; $V(0)$ via `gamma_corr_coulomb_at_zero`
	`gamma_corr_response`	dipoles, Y, X, B, E; $S_{\alpha\beta}(\mathrm{i}\omega)$, $T_{P\alpha}(\mathrm{i}\omega)$
	`gamma_corr_tabulate`	2D: $K_{\mathrm{h}}(\mathbf{R})$, $K_{\mathrm{w}}(\mathbf{R})$ tables; 3D: `gamma_corr_direction_coefficients`
`gw_compute_V_and_W`	`gamma_corr_subtract_divergent_head`	2D only: $V(0) - D_{\mathrm{box}} c c^{\mathrm{T}}$, Eq. (23)
`compute_W_MIC`	`gamma_corr_add_to_W`	2D: $W^{\mathrm{MIC}}_{PQ}(\mathrm{i}\tau)$ += $\Delta W_{PQ}(\mathrm{i}\tau)$, Eq. (45)
Γ driver, after Σ^c	`gamma_corr_add_to_sigma`	3D: $\Sigma^c_{\mu\nu}(\pm\mathrm{i}\lvert\tau\rvert)$ += $\Delta\Sigma^c_{\mu\nu}$, Eq. (73)

Γ. The full-k gap converges from below towards the Γ-only value; at 24×24 the absolute quasiparticle energies at Q agree to 16 meV (valence) and 24 meV (conduction), with identical Σ^x and v^{xc}, and the gap at P to 56 meV. The larger Γ-only cells sample a different point nearest to K (13/40 and 19/56 instead of 11/32) and move the gap by 11 and 34 meV; the full-k gap itself differs by 37 meV between K and P, so these are sampling shifts rather than cell convergence. With the standard Padé continuation of our earlier builds the same comparison gave 5 and 14 meV at Q and 20 meV at P (Table V); the averaged continuation amplifies the small Σ^c differences between the folded supercell spectrum and the primitive-cell spectrum (2.043 vs 2.082 eV at the valence state of Q at the same 8.7 Å exchange cutoff).

TABLE II. h-BN monolayer, aug-SZV tier-1 basis: direct G_0W_0 gap at P and quasiparticle energies at Q (eV); one master build, averaged Padé continuation in both codes. Γ-only: 8×8 supercell with the head-and-wing correction (the 10×10 and 14×14 entries are the gap at the sampled point nearest to K of those cells, 13/40 and 19/56). PR: primitive cell, $N\times N$ mesh.

	gap(P)	VB(Q)	CB(Q)	Σ^c VB / CB
PR 6×6	6.714	−5.013	2.169	2.050 / −3.088
PR 8×8	6.780	−5.051	2.161	2.082 / −3.113
PR 12×12	6.801	−5.082	2.143	2.118 / −3.146
PR 16×16	6.804	−5.096	2.129	2.134 / −3.161
PR 20×20	6.807	−5.105	2.122	2.142 / −3.168
PR 24×24	6.810	−5.112	2.119	2.146 / −3.172
Γ-only 8×8	6.866	−5.128	2.143	2.043 / −3.148
Γ-only 10×10	6.855			
Γ-only 14×14	6.832			

B. 2D benchmark with a DZVP basis

Four planar hexagonal monolayers (h-BN, SiC, h-GaN, h-AlN) with DZVP-MOLOPT-SR-GTH orbital bases and MOLOPT RI sets at the $2\cdot10^{-3}$ fit-error level, 15 Å vacuum, cutoff 500 Ry, 10 minimax points, `CUTOFF_RADIUS_RI` 5 Å, `SIZE_LATTICE_SUM` 4. The Γ-only cells are 9×9 (162 atoms; the folded manifold contains K) and 12×12 (288 atoms); the PR meshes contain K as well. Table III lists the indirect gap over the sampled points. The PR series is converged to 5 meV between 30×30 and 36×36, the Γ-only results to 3–15 meV between 9×9 and 12×12. Three of the four materials agree with the converged reference to 22–74 meV, and their Hartree–Fock gaps to 0.01–0.05 eV (h-BN 13.267 vs 13.278, SiC 8.399 vs 8.373, h-GaN 9.733 vs 9.779 eV). h-AlN deviates by 0.28 eV in the G_0W_0 gap and by 0.13 eV in the Hartree–Fock gap. State by state (9×9 against PR 36×36) the difference is spread over four contributions of the same sign: Σ^x at the valence maximum at K (−19.081 vs −19.149 eV) and at the nearly-free-electron conduction minimum at Γ (−4.894 vs −4.834 eV), together 0.13 eV, and Σ^c at the valence maximum (2.463 vs 2.414 eV) and at the conduction minimum (−2.448 vs −2.347 eV), together 0.15 eV; $\varepsilon^{\mathrm{DFT}}$ and v^{xc} of the four states are identical to the meV. Neither grid is responsible. The PR grid is converged for AlN (`RS_AO_RATIO` 20 gives 6.318 eV against 6.319 eV at the default ratio 7), and our Σ^x is converged with the truncated-Coulomb radius (12.1 Å at 9×9, 16.2 Å at 12×12: 4 meV). On our side the exact-RI tensor driver with the same Γ-only W and the same correction (Sec. IV A) reproduces the RI-RS result of the 6×6 AlN cell to 1 meV in every occupied and low-lying virtual state: gap 6.241 eV in both, and $\varepsilon^{\mathrm{DFT}}$, Σ^c, Σ^x, v^{xc} of the two band edges identical to the printed meV (−19.063 and −15.173 eV at the valence maximum, −4.901 and −8.832 eV at the conduction minimum for Σ^x and v^{xc}), at ten times the cost per χ time point (33 vs 3 s) and six times the memory. The remaining AlN difference therefore lies in the Γ-point treatment itself, or in the supercell-Γ versus full-k formulation (32–56 Å Coulomb truncation in the PR against 12–16 Å here, folded spectrum on the minimax grid here), and not in the numerics of either code; see Sec. VI C.

TABLE III. 2D benchmark, DZVP basis (a = 2.502, 3.090, 3.210, 3.110 Å): indirect G_0W_0 gap (eV) of the Γ-only supercell with the head-and-wing correction against the full-k PR at $N\times N$ meshes; one master build, averaged Padé continuation in both codes. Last column: Γ-only (9×9) minus PR 36×36.

system	9×9	12×12	PR 12	PR 18	PR 24	PR 30	PR 36	Δ
h-BN	7.837	7.834	7.827	7.839	7.847	7.854	7.859	−0.022
SiC	4.224	4.221	4.261	4.270	4.273	4.275	4.278	−0.054
h-GaN	4.464	4.479	4.514	4.527	4.532	4.535	4.538	−0.074
h-AlN	6.299	6.303	6.541	6.555	6.565	6.571	6.576	−0.277

C. 3D: diamond

Diamond ($a = 3.567$ Å, primitive two-atom cell, SZV-MOLOPT-SR-GTH, RI set `RI_20_2_2_1_1_0_0_0`, cutoff 400 Ry, 10 minimax points, `SIZE_LATTICE_SUM` 2, RI-metric cutoff 3.0 Å, Σ^x truncated at half the cell height) in $n \times n \times n$ supercells of 54, 128 and 250 atoms (the 432-atom cell was abandoned after seven hours on four nodes, still in the fit stage). Table IV shows the indirect G_0W_0 gap over the folded Γ manifold for the successive stages of the bulk correction, together with the PR at $N \times N \times N$ meshes and the Hartree–Fock gaps. Two things are established independently of the PR. First, the exchange: the exact-RI tensor implementation of the same 128-atom cell gives $\Sigma^x = -18.957$ eV (valence triplet) and -9.767 eV (conduction) and a Hartree–Fock gap of 15.125 eV, identical to the RI-RS values ($-18.957/-9.767$, 15.123 eV), whereas the PR at $4 \times 4 \times 4$ is 0.4–0.6 eV off in Σ^x and splits the Γ triplet by 0.2 eV, reaching the correct exchange only at $8 \times 8 \times 8$. Second, the structure of the correction: the head alone shifts the valence and conduction edges by $+1.58$ and -1.80 eV at $3 \times 3 \times 3$ (the $\pm w/2$ of a uniform potential with $w = -3.5$ eV, $\varepsilon_\infty = 3.05$); the head-dipole term adds $+0.20$ eV to the gap at 54 atoms and $+0.06$ eV at 128 atoms, the wing -0.03 and -0.00 eV, and the wing-wing term -0.002 eV, i.e. the dipole terms scale as $1/\Omega$ as they should. The RI-space application of the same head, by contrast, gave -96 eV. Degeneracies of the folded manifold are preserved to 3 meV. With the full kernel and the one master build the Γ-only gap runs $9.210 \to 9.970 \to 10.100$ eV (54, 128, 250 atoms; the last step $+0.13$ eV), the PR $9.239 \to 9.519 \to 9.593 \to 9.553 \to 9.565$ eV (4^3 to 12^3 meshes), converged to 0.01 eV from 8^3 on. The Hartree–Fock gaps rise on both sides towards the same value, 15.216 eV at 250 atoms and 15.120 eV at 12^3 with $+0.04$ eV per step left in the PR, so the exchange is consistent to 0.1 eV while the G_0W_0 gap of the Γ-only cell lies 0.5 eV above the converged full-k value at 250 atoms. The whole of that difference is in Σ^c.

Two further checks bound the numerics on our side. First, the exact-RI tensor driver with the same Γ-only W and the same orbital-basis correction (Sec. IV A) was run on the 54- and 128-atom cells. On the 128-atom cell it gives the same gap as the RI-RS code, 9.970 eV, and $\varepsilon^{\mathrm{DFT}}$, Σ^c, Σ^x, v^{xc} of every state of the valence triplet and of the conduction minimum identical to the printed meV ($1.950/1.932/1.932$, -18.957, -17.093 eV and -3.206, -9.767, -16.568 eV). On the 54-atom cell (standard-Padé builds) its gap is 9.386 eV against 9.039 eV from the RI-RS grid, but the difference is almost entirely exchange: the Hartree–Fock gaps are 15.074 and 14.838 eV, the RI-RS Σ^x is too negative by 0.06 eV on average over the occupied states and by 0.14 eV over the lowest virtuals, whereas Σ^c at the two band edges agrees to 0.004 and 0.036 eV. The $3 \times 3 \times 3$ cell truncates the Coulomb operator of Σ^x at 3.1 Å, the same length as the 3.0 Å RI-metric cutoff that defines the fit, and the grid representation of such a short truncated operator is where the RI-RS fit is weakest; at $4 \times 4 \times 4$ (4.1 Å) the RI-RS and exact-RI Σ^x agree to 1 meV, as stated above. The 54-atom RI-RS entries of Table IV are therefore grid-limited by ~ 0.2 eV, the larger cells are not, and the Γ-only result of the 128-atom cell, including the correction, is independent of the RI-RS fit. Second, the standard tensor k-point GW of CP2K on the primitive cell was attempted as an independent arbiter; with the 9 Å orbital and RI radii its real-space lattice sums (1607 cells of three-centre integrals, 12071 cells $\Delta\mathbf{R}$) cost 10.5 and 21.7 h per imaginary-time point on 8 ranks for the 6^3 and 8^3 meshes, the code has no χ restart, and the runs were abandoned at the 24 h limit. The 0.5 eV between the two Γ-point treatments at convergence is thus not settled by a third code; what is settled is that it is not a fit error of either.

TABLE IV. Diamond, SZV basis: indirect G_0W_0 gap over the folded Γ manifold (eV) for the stages of the bulk correction (columns none to full: development builds with the standard Padé continuation), the full kernel with the one master build and the averaged Padé continuation (full*), and the full-k PR from the same master build. HF: Hartree–Fock gap with the truncated Coulomb operator of each run (master build).

cell	atoms	none	head	+h.-dip.	full	full*	HF
3^3	54	12.249	8.871	9.066	9.041	9.210	14.833
3^3, exact RI	54				9.386		15.074
4^3	128		9.722		9.782	9.970	15.123
4^3, exact RI	128					9.970	15.125
5^3	250		9.910		9.934	10.100	15.216
PR 4^3						9.239	14.116
PR 6^3						9.519	14.784
PR 8^3						9.593	15.011
PR 10^3						9.553	15.077
PR 12^3						9.565	15.120

VI. DISCUSSION

A. Why the old k-mesh large-cell scheme is not a reference

The large-cell Γ-point GW implementations in CP2K (`tensor_large_cell_Gamma`, `ri_rs_large_cell_Gamma`) build the screened interaction of a 2D supercell on a k-mesh `KPOINTS_W` $nn1$ of the supercell's small Brillouin zone, with a second "extrapolation" mesh of $2n$ points and a Coulomb matrix $V(\mathbf{q})$ from a minimum-image truncated real-space lattice sum. Band gaps from this scheme were used as the reference against which the Γ-only scheme was judged, and the two differ by 0.12 eV on the largest cell we can afford. The old scheme carries the systematic error; the error is the spurious net charge of RI-fitted pair densities identified by J. Wilhelm in CP2K PR #6447 [2]; and the Γ-only numbers are confirmed by his full-k-point code to 15–60 meV (Sec. V A). Numbers that came out of the old scheme for 2D systems should therefore not be used as references.

a. The observation. Monolayer h-BN, $a = 2.502$ Å, 15 Å vacuum, PBE, aug-SZV-MOLOPT-GTH tier-1 orbital and RI bases, 10 minimax points, `CUTOFF_RADIUS_RI` 5 Å, identical in every code and run. "Gap" is the direct G_0W_0 gap at the k-point $P = (11/32, 11/32)$ of the primitive zone, the

point nearest to K that an 8×8 supercell with a 2×2 DOS mesh samples (Table V).

TABLE V. Direct G_0W_0 gap of monolayer h-BN at $P=(11/32,11/32)$ from the old large-cell scheme, the Γ-only scheme of this work, and the full-k RI-RS GW of PR #6447 (primitive cell). All entries of this table are from the development builds with the standard Padé continuation, which is the only way to compare with the old scheme; the one-build numbers are in Table II.

scheme	cell / mesh	gap (eV)
old, `KPOINTS_W 4 4 1` + extrap.	8×8 supercell	8.152
old, `KPOINTS_W 4 4 1` + extrap.	14×14 supercell	6.582
old, `KPOINTS_W 2 2 1` + extrap.	14×14 supercell	6.595
Γ-only + head/wing (this work)	8×8 supercell	6.710
Γ-only + head/wing	10×10 supercell	6.705
Γ-only + head/wing	14×14 supercell	6.699
full-k RI-RS, PR #6447	6×6 mesh	6.590
full-k RI-RS, PR #6447	8×8 mesh	6.658
full-k RI-RS, PR #6447	12×12 mesh	6.681
full-k RI-RS, PR #6447	16×16 mesh	6.684
full-k RI-RS, PR #6447	20×20 mesh	6.687
full-k RI-RS, PR #6447	24×24 mesh	6.690

Three things stand out. (i) The old scheme is not converged with the cell: 8.15 eV at 8×8 against 6.58 eV at 14×14, and an earlier sweep with the tier-2 basis (11.85, 9.74, 9.66, 9.43, 8.63, 7.54 eV for $n=5,6,7,8,10,12$) was still falling at 30 Å. (ii) The Γ-only scheme is converged to 5 meV in the gap from 8×8 on. (iii) The independent full-k-point code converges from below onto the Γ-only value; state by state at the supercell Γ point the absolute G_0W_0 energies agree to 5 meV (valence) and 14 meV (conduction) with the 24×24 mesh, with identical Σ^x and v^{xc} and Σ^c within 10 meV. The old scheme sits 0.12 eV below both at 14×14 and is still drifting downwards, so the 0.12 eV is its error, not the new code's.

b. The mechanism: a spurious charge in the fitted pair densities. In RI-RS a product of two orbitals is interpolated from its values on the atom-centred grid points $\mathbf{r}_\ell$ and fitted to the auxiliary basis through the fit coefficients $Z_{\ell P}$,

$$\begin{aligned}\psi_i(\mathbf{r})\psi_a(\mathbf{r}) &\approx \sum_P \Big[\sum_\ell Z_{\ell P}\,\psi_i(\mathbf{r}_\ell)\psi_a(\mathbf{r}_\ell)\Big]\phi_P(\mathbf{r})\\ &\equiv \sum_P d_{ia,P}\,\phi_P(\mathbf{r}).\end{aligned}\tag{75}$$

The exact transition density is neutral, $\int\psi_i\psi_a=0$ by orthogonality. The fitted one is not: its charge $m_{ia}=\sum_P c_P d_{ia,P}$ is a small but nonzero number set by the grid and the fit (the grid cannot represent orthogonality, and the fit is regularised). This was found by J. Wilhelm while converging his full-k-point RI-RS GW.

A spurious charge is harmless for a molecule and nearly harmless for a 3D crystal, but not in two dimensions. On the $\mathbf{q}$ mesh of a 2D calculation the Coulomb kernel in the charge channel is $v(\mathbf{q})=2\pi/(Aq)$, so the charge mode of χ enters the dielectric function as

$$\varepsilon_{00}(\mathbf{q})-1 \;\sim\; \frac{2\pi}{Aq}\sum_{ia}\frac{2\,|m_{ia}|^2\,\Delta_{ia}}{\Delta_{ia}^2+\omega^2} \;=\; \frac{C}{q},\tag{76}$$

a $1/q$ divergence with a spurious prefactor. Summed over a 2D mesh the mesh points approach $\mathbf{q}=0$ as $1/N$, and $\sum_{\mathbf{q}}1/q$ over an $N\times N$ mesh grows like $\ln N$: the screening, and with it the gap, drifts logarithmically with the mesh instead of converging. In Wilhelm's calculations the WSe_2 gap moved from 2.222 to 2.197 eV with the mesh; after projecting the charge out of every fitted density, $d'=d-c\,(c^\dagger d)/(c^\dagger c)$, it is 2.233 eV and stable; the 2D h-BN regression reference moved by +0.92 eV.

The old large-cell scheme has the same fitted densities with the same spurious charge, but its Coulomb matrix is not $2\pi/(Aq)$. $V(\mathbf{q})$ is a real-space lattice sum truncated at the minimum-image radius $R_c\approx L/2$ of the supercell, whose charge-channel value is finite, $\int_0^{R_c}2\pi r\,dr/r=2\pi R_c$ per cell area. The leak therefore does not diverge; it produces a finite spurious screening proportional to $|m|^2R_c/A\propto1/L$, i.e. a uniform downward shift of Σ^c that decays only as the inverse cell size. That is exactly the signature in Table V: a gap that is too small, converges slowly from below, and is still moving at 30 Å. The sign of Wilhelm's fix (+0.92 eV for h-BN with his settings) and the sign of our discrepancy (old scheme below) agree.

c. Why the Γ-only scheme is immune. The Γ-only implementation never lets a fitted density interact through the charge channel of a sampled $V(\mathbf{q})$:

- the body of the dielectric matrix of the correction is built with the neutral projector, $V_b=\Pi V(0)\Pi$, Eq. (30), so the charge direction of the auxiliary basis carries no Coulomb interaction at all in that body;
- the head and wings of ε in the charge channel are built from the exact velocity-gauge transition dipoles, Eq. (32), checked by the sum rule of Eq. (33), and integrated analytically over the whole small Brillouin zone with the $2\pi/q$ kernel;
- the regular $W(\Gamma)$ uses $V(0)$ with only the divergent $\mathbf{G}=0$ constant removed, Eq. (23), and the charge direction of that matrix is quenched in ε^{-1}.

This is the same treatment of the charge channel as Wilhelm's Γ model, which is why the two codes agree once his leak is removed. The agreement (15 meV at 8×8 versus 24×24, with the remaining difference of the size expected from the 8×8 cell) is the actual validation of the Γ-only scheme; the old large-cell numbers are not.

d. Consequences. Band gaps of 2D systems produced by `tensor_large_cell_Gamma` or `ri_rs_large_cell_Gamma` with `KPOINTS_W` $nn1$ ($n>1$) carry a cell-size-dependent error of order 0.1 eV at 30 Å and more than 1 eV on cells of 20 Å, and should not be used as references or regression values. For 2D the large-cell code should be run Γ-only (`KPOINTS_W 1 1 1`), which activates the head-and-wing correction; the same holds for 3D cells (`PERIODIC XYZ`), where $V(0)$ is an Ewald sum with the $\mathbf{G}=0$ term omitted.

B. Gaussian form factors of the auxiliary charges are not needed

The 2D correction treats the auxiliary Gaussians as point charges at their centres, the limit $q_{\max}\sigma_P \ll 1$ of Eq. (29); the Gaussian form factors $e^{-q^2/4\alpha}$ are neglected. With the most diffuse s exponent of the aug-SZV RI basis ($\alpha = 0.199$) the head kernel changes by 7 % on the 3×3 cell, 1.8 % at 6×6, 1.1 % at 8×8 and 0.4 % at 14×14. On a head contribution of ~ 30 meV to the gap this is below 1 meV on production cells, and the form factors are not implemented. In 3D the question does not arise: the correction is applied in the orbital basis, Eq. (65), and involves no auxiliary charges.

C. Remaining approximations

Four approximations of the model remain. (i) The body of W is taken at Γ only, so its dispersion across the small Brillouin zone is not corrected; this error decreases with the cell size, and the cells for which the method is intended have a flat body. (ii) In 2D, the constant-function head carries no $e^{-q|z-z'|}$ form factor for the finite thickness of the layer: a state that extends several Å into the vacuum couples to the charge mode with a factor $e^{-q|z|}$ that is 0.7 at $q = 0.1$ Å^{-1} and $z = 4$ Å. This was the first candidate for the AlN difference of Sec. V B, but it does not survive the cell test: the missing factor is first order in q and its effect would shrink with the radius of the small Brillouin zone, by about one half from the 9×9 to the 12×12 cell, whereas the gap moves by 4 meV; and the exact-RI tensor driver reproduces the RI-RS value, so no fit error is involved either. Our integral runs over the whole small Brillouin zone ($q \le \pi/9a$ for a 9×9 cell) while the PR applies its model only to the cells adjacent to Γ ($q \le \pi/36a$); which of the two Γ-point treatments is closer to the full-k limit for a diffuse state is not settled by the present data. The difference is not a property of the box of the lattice sum: the subtracted constant D_{box} grows with the box while the finite remainder of the charge channel does not (h-BN, 8×8: $s = 4$ and $s = 5$ boxes differ by 1 meV in the gap). (iii) In 3D, the small-q expansion of the pair phase, Eq. (63), holds for $|\mathbf{q}\cdot\mathbf{d}_{nm}| < 1$; nearly degenerate pairs with a finite coupling violate it, and their dipoles are clipped at $1/q_{\max}$ (484 of 512^2 pairs in the 128-atom diamond cell). (iv) The orbital-diagonal application neglects the off-diagonal elements $\Delta\Sigma^c_{nm}$, $n \ne m$, which only matter if the correction were allowed to mix quasiparticle states.

VII. CONCLUSIONS

Γ-only RI-RS *GW* of periodic large cells is made well defined by separating the charge channel of the auxiliary basis. In 2D slabs its divergent lattice sum is removed from $V(0)$ as a single rank-one constant, $D_{\text{box}}\,cc^{\mathrm{T}}$, computed exactly as a boundary integral of the image box; in 3D bulk cells $V(0)$ is the Ewald sum with the $\mathbf{G} = 0$ term omitted. The charge channel is put back analytically through the normalized constant function, whose screened head and wings follow in closed form from the velocity-gauge transition dipoles and the Gaussian dielectric body and are integrated over the whole small Brillouin zone of the supercell; nothing is subtracted. In 2D the correction is applied in the auxiliary basis, at negligible cost. In 3D, where the head is eV-sized and an RI-space application is amplified by the fit, it is applied to Σ^c in the orbital basis through a single orbital-pair kernel that contains the head, the head-dipole and the wing terms. The implementation in CP2K is activated automatically by `KPOINTS_W 1 1 1` for `PERIODIC XY` and `PERIODIC XYZ`, and chooses replicated or distributed dense algebra from a memory estimate. The earlier k-mesh large-cell scheme carries a spurious charge leak of the fitted pair densities and should not be used as a reference for 2D systems. Numerically, with one master build on both sides, the slab correction reproduces the converged full-k RI-RS *GW* of PR #6447 to 20–75 meV for h-BN (two bases), SiC and h-GaN, with a 0.28 eV deviation for h-AlN that the exact-RI tensor driver reproduces to 1 meV, so it is a property of the Γ-point models and not of the RI-RS fit. For diamond the Γ-only RI-RS result of the 128-atom cell, exchange, correlation and correction, is identical to the exact-RI tensor result to the meV, and the dipole terms of the bulk kernel scale as $1/\Omega$ as derived; but the gap is still rising by 0.13 eV from 128 to 250 atoms and lies 0.5 eV above the converged full-k value there, entirely in Σ^c. The bulk Γ-point model is therefore validated as an implementation and not yet as a reference: which of the two Γ-point treatments is closer to the full-k limit remains open, and the standard tensor k-point reference proved too costly to decide it. The correction costs about 1 % of a 128-atom run.

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The full-k reference calculations use the implementation of J. Wilhelm (cp2k PR #6447). Computations were carried out on the Noctua 2 cluster of the Paderborn Center for Parallel Computing (PC^2).

[1] J. Wilhelm, *Periodic real-space RI for full-k GW band structures*, manuscript in preparation.

[2] CP2K pull request #6447, https://github.com/cp2k/cp2k/pull/6447.